Detection of Hair-line Cracks in IITJ Buildings Using Ground Penetrating Radar (GPR)

A Project Report Submitted by
Siddharth Chauhan
(B21CI041)

in partial fulfillment of the requirements for the award of the degree of
Bachelor of Technology

Under the Supervision Of
Dr. Pradeep Dammala

Indian Institute of Technology, Jodhpur
Department of Civil and Infrastructure Engineering
November 2024

Declaration

I, hereby declare that the work presented in this Project Report, titled Detection of Hair-line Cracks Using Ground Penetrating Radar (GPR) submitted to the Indian Institute of Technology Jodhpur in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology is a bonafide record of the research work carried out under the supervision of Dr. Pradeep Dammala. The contents of this B.Tech. I hereby declare that this Project Report, in whole or in part, has not been, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Siddharth Chauhan
B21CI041

Certificate

This is to certify that the Project Report titled Detection of Hair-line Cracks Using Ground Penetrating Radar (GPR), submitted by Siddharth Chauhan (B21CI041) to the Indian Institute of Technology Jodhpur for the award of the degree of B.Tech. is a bona fide record of the research work done by him under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Dr. Pradeep Kumar Dammala

Acknowledgements

I would like to express my gratitude to the several individuals who helped enable this project. Firstly, I sincerely thank my project supervisor, Dr. Pradeep Dammala, Professor of Civil Engineering at Indian Institute of Technology, Jodhpur, for his invaluable guidance, feedback, and support throughout the duration of the project. I also thank Bharat Solanki, Junior Technical Superintendent at the Department of Civil and Infrastructure Engineering, Indian Institute of Technology, Jodhpur, for his support throughout the duration of the project.

I am grateful for the insights provided by various academic works and technical papers cited in the references, which tremendously benefited my literature survey and understanding on this topic.

It has been a great learning experience for me to apply surveying concepts I studied to study real-life infrastructure challenges.

Siddharth Chauhan
BTech (Civil Engineering)
Indian Institute of Technology, Jodhpur

Content

Problem Statement

There is persistent seepage in the buildings of IIT Jodhpur during the rainy season that is threatening the structural and material durability. Even though cracks appear on some surfaces, they can't be responsible for the seepage as the small cracks are hidden or tiny pores. The traditional visual inspection cannot detect the tiny defects below the surfaces; hence, a more accurate non-destructive technique is needed. The GPR technology is to be applied to inspect the building structures of IIT Jodhpur in a manner so that both deductible and hidden cracks related to moisture seepage can be observed and some outlook for efficient maintenance and remediation strategies comes to view.

Abstract

A hairline crack detection in concrete structures is quite crucial for structural integrity and preventing water penetration-often leading to long-term degradation. This study proposes Ground Penetrating Radar (GPR) 1600 MHz antenna for crack identification at the IITJ buildings' walls, paying attention to potential seepage by water during the wet season. Data were therefore collected on various buildings such as E-5 Berm, G-3 Hostel that comprises terrace and rooms, Y4 Hostel outer side, and O4 Hostel outer side. Data analyzed by the help of RADAN Software. It is clear that this GPR model could identify only visible cracks because of its frequency of wavelength that was limited to discover the small cracks not visible to the naked eye.

It is noted that the outer walls of such structures are made from Jodhpur sandstone on the outer surface; then there is insulation, followed by inner brickwork. This high rate of water absorption in sandstone combined with its long-term retention of moisture along with limited exposure to sunlight seems to allow the penetration of water to the next level into the polystyrene foam (thermal insulation material) covering the entire brick wall, leading to internal seepage. Notably, the GPR system detected larger cracks effectively but cannot detect minute or invisible cracks. This has an implication that water infiltration might also occur through micro-pores in the concrete or due to poor interlocking between the joints in some precast structures. These findings indicate potential applications of GPR in structural inspection at larger structural levels, and also serve as a suggestion that there is a greater need for concurrent methods that can identify micro-level cracks and pores to ensure effective mitigation of the seepage issues.

1. Introduction

Concrete structures, being integral parts of civil infrastructure, gradually deteriorate with time, as they get old, due to environmental factors and climate change effects. In particular, undetected hairline cracks in concrete can be a source of safety risks and compromise structural integrity, hence identification at early stages is very important to implement preventive maintenance and ensure long-term resilience. Between the available non-destructive testing methods, one method which has recently gained popularity for conducting minute discontinuities in concrete, such as hairline cracks without damaging the structure itself is Ground Penetrating Radar.

The principle working of GPR is that it radiates high-frequency electromagnetic waves into the concrete; these waves get reflected back upon finding any irregularity thus permitting detailed internal mapping. The travel time, amplitude and frequency changes are indicative of subsurface concrete conditions as they propagate through the medium. In recent years, the use of GPR became highly widespread due to the efficiency of the technique at inspecting large areas much more quickly than more invasive methods that take a lot longer.

This technical effort deals with safety in concrete structures of IIT Jodhpur with an early indication of hairline cracking. In the current study, GPR is employed to detect micro-cracks in IITJ buildings, which shall improve safety and structural integrity. This paper targets achieving accuracy in internal cracks using high-frequency antennas that may not be present within the structure as it is visually apparent on the surface, leading to adaptive maintenance strategies. It will contribute to preserving the structural integrity of the building since relevant problem zones will be identified along with guiding timely repairs.

2. Literature review

GPR has been utilized in NDT for concrete structures as it evaluates the integrity of such structures, detects cracks and voids, and other anomalies in their structure. GPR sends high-frequency electromagnetic waves into the concrete and records the reflections from the different materials and defects [1]. It is an important tool for identifying hairline cracks and in assessing the condition of concrete, especially historical and reinforced concrete structures.

Figure 1: *GPR antenna and controller.*

2.1. Working Principle of GPR

Ground Penetrating Radar (GPR) is a technique based on the principle of transmitting high-frequency electromagnetic (EM) waves into a material and analyzing reflected signals that unveil subsurface characteristics. The most fundamental components of the GPR system include a transmitter, receiver, and control unit, which are responsible for signal generation, timing, and gathering data. The subsurface is bombarded by EM pulses emitted by the transmitter antenna. Along a path through the soil, the waves can be reflected back to the receiver antenna whenever they encounter the interface between two materials with different dielectric properties. Any change in the dielectric constant among different materials will modify the velocity of the EM wave and create what is called unique reflections captured by the receiver antenna. The receiver records these backscattered signals with signatures of both time delay and amplitude, through which depth and composition of buried features are estimated [12].

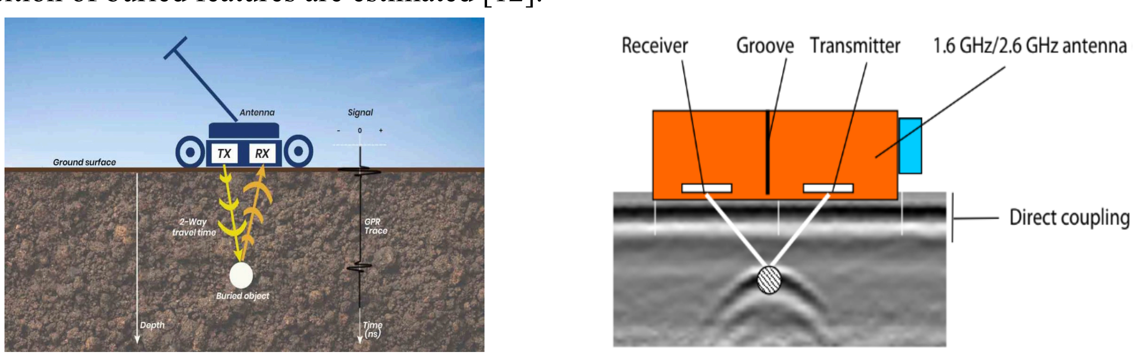
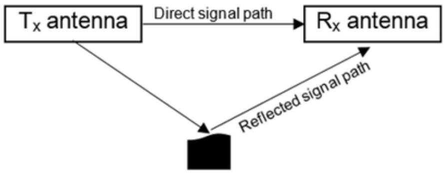
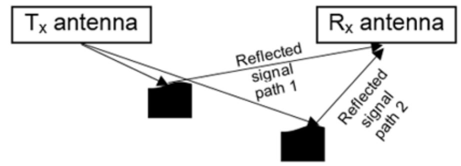


Figure 2: *GPR antenna locating a target* [13].



$$\Delta t = \frac{\text{Direct signal path} - \text{Reflected signal path}}{V} = \frac{\Delta\text{Length}}{V}$$

Figure 3: *Transmitter blanking that occurs because of overlapping between direct signal and reflected signal.*



$$\Delta t = \frac{\text{Reflected signal path 2} - \text{Reflected signal path 1}}{V} = \frac{\Delta\text{Length}}{V}$$

Figure 4: *Transmitter blanking that occurs because overlapping between two reflected signals that have a similar path length.*

2.2. Applications of GPR in Concrete Crack Detection

GPR can be used to recognize cracks in concrete and in identifying cracks with higher amplitude and lower frequency, surface and subsurface of the same concrete. It has been shown that GPR effectively identifies visible cracks but fails to identify hairline cracks because the identification depends on the frequency of the antenna used [2]. High-frequency antennas provide higher resolution but reduce penetration depth, which is useful in shallow crack detection. Low frequency antennas give higher penetration but lesser resolution [3].

In addition GPR sensitivity on hairline crack is dependent on factors of scan spacing, the speed of the survey, and frequency of the antenna [4]. The study underlined stress on optimizing all the above parameters to increase the accuracy of crack detection. Since structural damage can be great because of hairline cracks, their early detection has to be conducted in order to avoid degradation in the structure [5].

2.3. Scan Spacing and Data Integrity

As noted in the **Geophysical Survey Systems Inc**. (2017) handbook, scan spacing determines the resolution and quality of GPR data, translating into how easily small cracks or voids are recognized at high scan spacing values. It may take around 10 scans/inch in smaller defects, which is 5 scans/cm. The project also utilized scan spacing setup to maximize hairline cracks capturing on IITJ buildings. There is scientific basis that indicates that the increases in scan density result in more detailed images, hence a better recognition of the details of a structure, and thereby yield a more accurate structural assessment [6]. However, it also means that survey speed will be lower when higher scan density is used.

2.4. Techniques to calibrate depth and signal processing

Depth calibration would also be another critical step in GPR analysis that ensures accurate measurements of crack depths. As illustrated by studies in this area, utilization of concrete dielectric constants or ground truth, where there is known target depth, increases the accuracy of measured depth [7]. In the present research work, GPR data acquired from structures of IITJ have been

analyzed to detect crack depth and other anomalies by the use of RADAN software. The depth scale was calibrated using dielectric tables and signal velocity computations, enabling accurate localization of structural elements including cracks and reinforcement [8].
Another common processing step often performed on GPR data is hyperbolic shape analysis known as migration. This processing technique enhances resolution by converting hyperbolic reflections into point sources, which simplifies the interpretation of the acquired data. Migration has been proven to be a practical clutter-reduction tool, making it easier to identify weak signals [9]. Migration was also used to process the GPR data to reveal hairline cracks in this project, thus showing that the previous claims stating that GPR is capable of detecting small variations in structures are valid.

2.5. Hairline Crack Detection Limitations

While GPR has numerous advantages, the detection of hairline cracks beyond the resolution limit of GPR systems is difficult. Despite GPR's capability to detect cracks as small as 6mm, the less than this size are quite often not visible because of the limitations in the frequency range used by the antennas and scan spacing [10]. This also causes a problem for our study context because sometimes hairline cracks that are hardly even visually noticeable go unnoticed by GPR, particularly on tight reinforcement or complex structures regions. Apart from these, other factors such as moisture content and surface roughness outside may influence the wave and thereby make the detection even more complicated .

2.6. Future Development on GPR Technology and Prospects

The trend in GPR technology over the last few years has been impressive. Among the latest developments include multi-frequency antennas, 3-D GPR imaging, and automated data processing algorithms that have considerably raised the capacities in GPR to detect small faults and to have an all-round assessment of concrete structures. Research conducted by Han et al. illustrated how the integration of GPR with AI and ML was able to enhance crack detection accuracy and efficiency through automation and examination using large datasets. Such approaches seem very promising for future applications of GPR in the context of building inspections and structural health monitoring.

3. Objective

The purpose of this project is to use the Ground Penetrating Radar model SIR 4000 having an antenna of 1600 MHz for the detection of hairline cracks and moisture zones in the concrete structures at IIT Jodhpur. In the light of established studies in non-destructive testing (NDT), this paper aims to assess the GPR in terms of identifying surface irregularities along with the limitations regarding the minute defects that could be a cause for the seepage and the deterioration of the structures. Finally, these assessments point out how GPR applications can be made better for concrete crack analysis.

4. Methodology

4.1. Study Area and Selection of Buildings

For the purpose of this research, I carried out GPR surveys on some buildings on the IITJ campus. The selected buildings that are included in the study are as follows:

- S-1 E-3 Berm
- S-2 G-3 Hostel (terrace and rooms)
- S-3 Y-4 Hostel (exterior)
- S-4 O-4 Hostel (exterior)

Figure 5: *Location of GPR survey points (S1-S4) within IIT Jodhpur campus master plan*

4.2. Data Collection Using GPR

As a part of the project, GPR with a 1600 MHz antenna was used to locate hairline cracks in several buildings of IIT Jodhpur. GPR surveys were carried out by transmitting high-frequency electromagnetic waves to the concrete structures through the GPR system. The system captures the reflected signals that reflect back from the concrete surface, and these are used for analysis of crack and void possibilities in the structures. It was aimed at detection of likely causes for experienced water ingress into the place during wet months, in view of probable crack development at the hairline and permeable wall structure. GPR scans are captured over areas identified by substantial subsurface reflection points to describe any discontinuity or cracking, as well as air voids inside the concrete wall at all the places being covered.

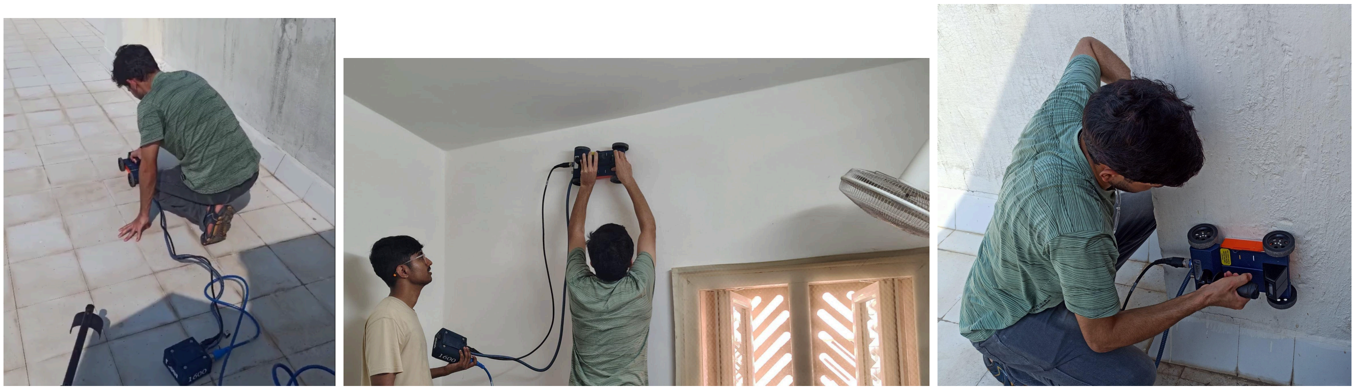


Figure 6: *Data collection using GPR within IIT Jodhpur campus*

At the time of the surveys, I used several parameters to maximize the collection of data, as given below:

- **Scan Spacing:** Several variations were performed in terms of scan spacing based on the required resolution. A higher scan spacing was applied to have a high possibility of detecting small cracks accurately.
- **Scan Frequency:** Several frequencies were tried, and high frequencies were only applied in cases where finer resolutions were required to identify very small hairline cracks. Lower frequencies, in any case, allowed penetration of the radar waves into the concrete to a greater extent.
- **Time of acquisition:** The speed at which the survey was taken was adjusted to ensure good quality data was collected without compromising on efficiency during the fieldwork.

4.3. Data Processing

The raw data collected during the data acquisition process were imported into the RADAN software for further post-processing operations. RADAN is a sophisticated tool that has been specifically designed for GPR data processing with the aim of enhancing the quality of the images to aid in the identification of cracks in the bonded structure. The workflow of data processing followed the following:

- **Colour Processing:** In the software, using the color table of RADAN, the 18 Dirt color setting is applied for better visibility. Through the Color Xform function, LM 8 selection is done to maximize the color contrast. The LM 8 and LM 4 settings especially improve scan clarity in concrete structures.
- **Time-zero correction:** The time-zero correction was applied to remove errors caused by air gaps between the ground surface and the GPR antenna cart.
- **IIR Filtering:** IIR filtering was applied for background noise elimination. The filter values for the high-pass were set at 90 MHz to 130 MHz, and the low-pass was set at 510 MHz, with a view to eliminating unwanted frequencies and clarity processes.
- **Noise Band Removal:** More noise reduction operations were performed by using the full scan band algorithm on bands that interfered with other quality signals.

- **Gain Adjust:** For improving the quality of an image, gain restoration and range gain adjustment were applied to increase the quality of an image. Specifically, the gain was exponential in form with eight gain points for enhancing the weak signal.
- **Hyperbola Fitting and Migration:** The ghost hyperbola was overlapped over the raw data to fit it with a hyperbola, while the alignment was optimized for proper locating of the target. Then, the migration procedure was carried out to erase both sides of the hyperbole while showing only the top part of the target in order to aid accurate crack locations.

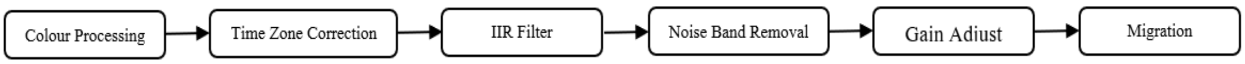


Figure 7: *Steps showing data processing*

4.4. Results and Interpretation

I Processed and analyzed the collected GPR data from various locations around the IITJ campus, namely S-1 (E-3 Berm), S-2 (G-3 Hostel terrace and rooms), S-3 (Y-4 Hostel exterior), and S-4 (O-4 Hostel exterior) using RADAN software to identify the subsurface abnormalities such as cracks, voids, and moisture zones:

- **S-1 (E-3 Berm):** There is a visible crack on this segment, which shows GPR to have the capability of detecting the surface cracks even though they are easy to see with the naked eye. The reflection pattern confirmed having a good dielectric contrast that reveals the depth and orientation of the crack inside the berm structure.

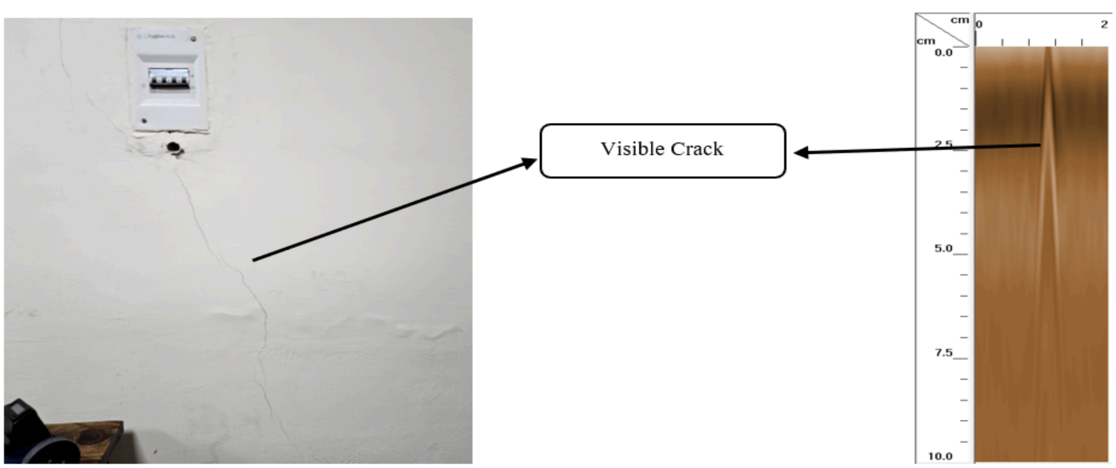


Figure 8: *Visible crack detection at E-3 berm.*

- **S-2 (G-3 Hostel terrace and Room 225):** In the G-3 Hostel's terrace, a damp zone or void zone was observed within the concrete at a reasonable depth. It can be presumed to be an indication of water penetration into the structure through different pathways.

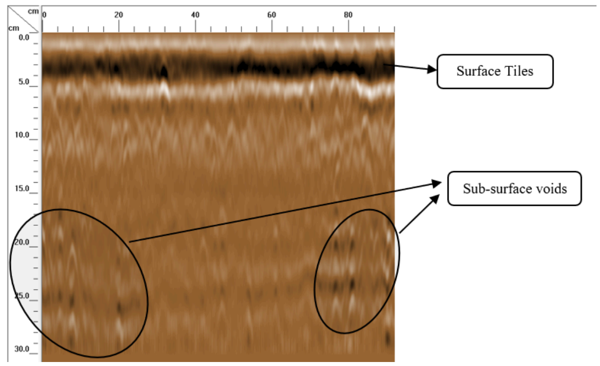


Figure 9: *Subsurface abnormalities detection at G-3 Hostel (terrace).*

In Room 225, reflections in GPR indicated the presence of damp zones or probable voids within the wall, which may be contributing to water seepage or weakness in the structure. Thereby, these are identified areas for future maintenance and repair.

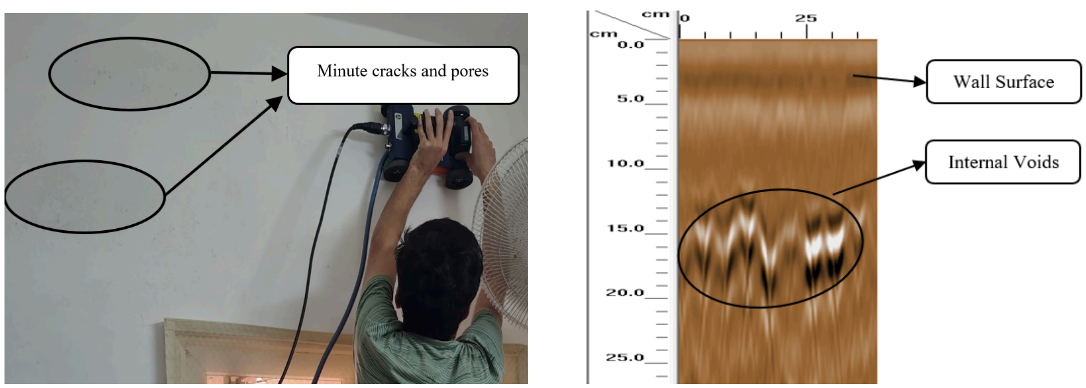


Figure 10: *Subsurface abnormalities detection at G-3 Hostel (room 225).*

- **S-3 (Y-4 Hostel exterior):** The GPR scanning successfully captured surface cracks and subsurface anomalies (voids, subsurface cracks and reinforcement) in the Y-4 Hostel exterior. In moist areas where no obvious cracks were visible, if there existed no visible cracks, the GPR generated no appreciable output because the minute pore sizes, sometimes even non-continuous pores, fell below its detection threshold.

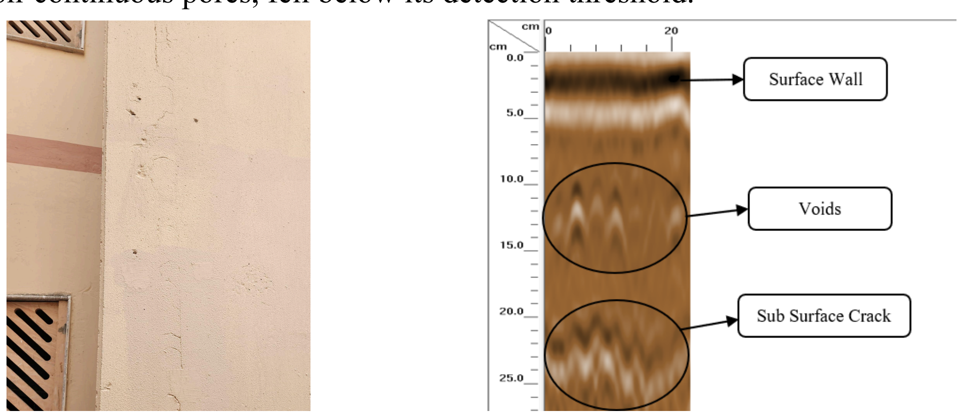


Figure 11: *Surface and subsurface abnormalities detection at Y-4 Hostel.*

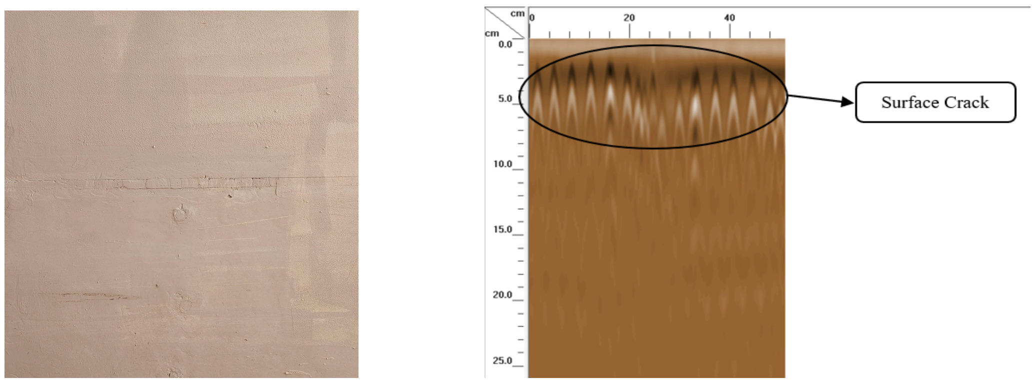


Figure 12: *Surface crack detection at Y-4 Hostel.*

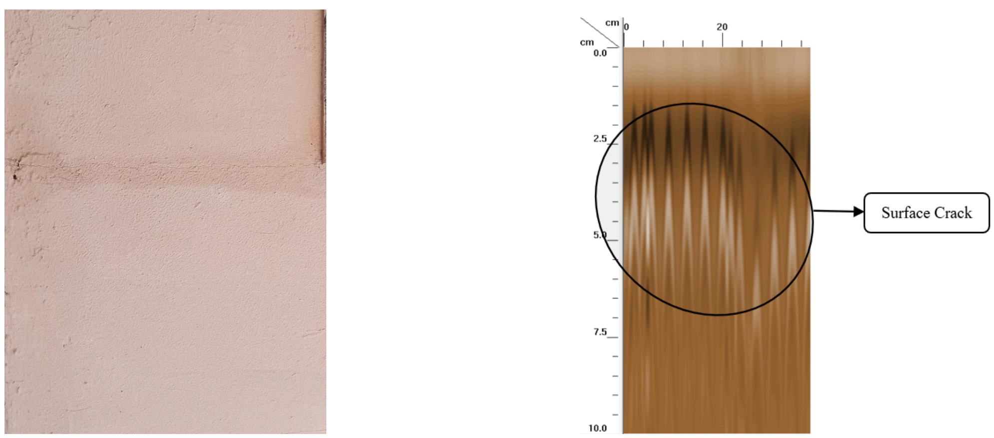


Figure 13: *Subsurface abnormalities (voids, subsurface cracks and reinforcement) detection at Y-4.*

- **S-4 (O-4 Hostel exterior):** The GPR scans at the exterior of the O-4 Hostel were very similar to those made outside Y-4. Its surface cracks, together with some subsurface anomalies, were recorded. Where there was moisture and no cracking was visible, GPR measurements were okay, which would mean possible microscopic pores that allow water penetration without a considerable dielectric contrast.

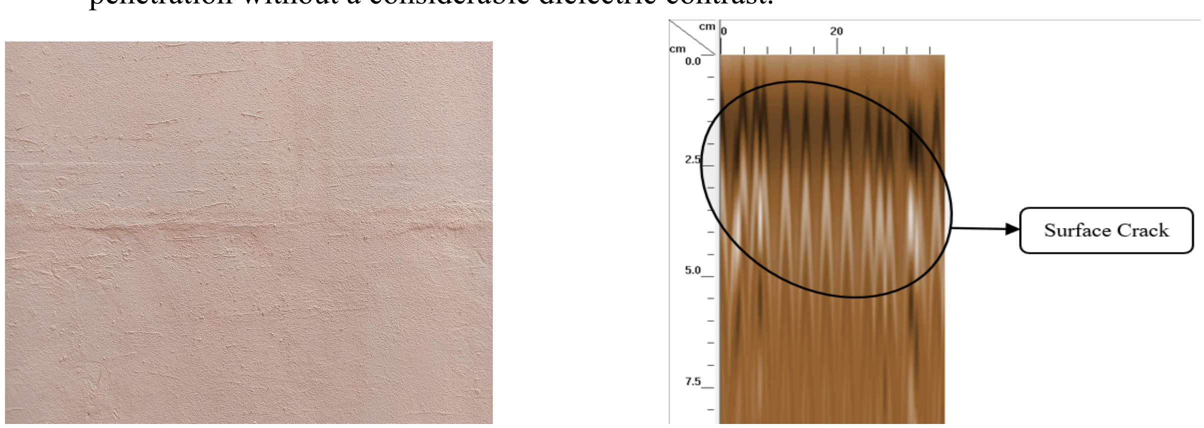


Figure 14: *Subsurface abnormalities (voids, subsurface cracks and reinforcement) detection at O-4.*

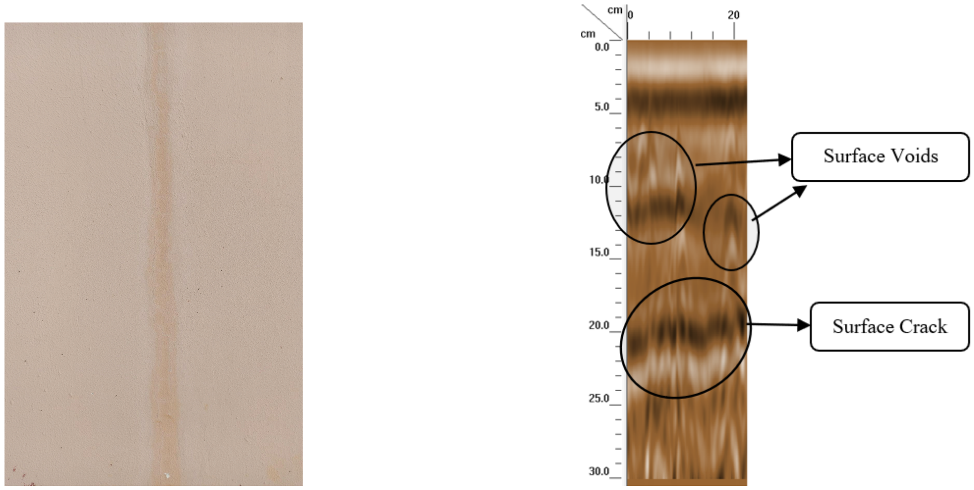


Figure 15: *Subsurface abnormalities (voids and subsurface cracks) detection at O-4.*

5. Conclusion

The project focused NDT for finding hairline cracks and possibly water-related problems in many of the buildings at IIT Jodhpur using GPR, taking note of areas that exhibit leakage due to rainwater during the rainy season. Several areas were covered: the E-3 Berm, the G-3 Hostel terrace and rooms, and the Y-4 and O-4 Hostel exteriors.

The cracks identified from the surface and subsurface according to GPR indicate areas where structural weaknesses are noted, particularly at critical structural weakness areas. It is well brought home, for example, with the successful implementation on the E-3 Berm and its environs in the Y-4 and O-4 Hostels with visible cracks detected. It then emerged that GPR was limited in areas prone to moisture where there were no cracks seen to exist and such places included specific spots on the G-3 Hostel terrace and Room 225 where zones of moisture or perhaps voids had been found, but a few mini-cracks or micro-pores through which water could seep were beyond the capability of the GPR to detect due to its frequency restraint since the GPR was incapable of detecting cracks less than 6 mm.

These findings imply that, although useful in detecting structural problems on large concrete areas, GPR does not always detect microscopic small defects that allow the water to seep in. In fact, this potential drawback forms a significant basis for using complementary methods in detecting possible minute concrete structure anomalies. It reiterates the possibility of GPR to evolve as a non-disruptive diagnostic tool for large structures and identifies areas where further inspection techniques may be developed and integrated to overcome the current limitations in the detection of micro-level damage.

6. Further Recommendations

To make the study more reliable and deeper in terms of findings, the following are some suggested improvements and future extensions:

6.1. Extend Data Collection Area

Extension of data collection area to other sections of the IITJ campus and associated structures would offer a comprehensive view of factors that may have led to water seepage and structural integrity issues. It would thus be possible to make comparisons across different building designs and their respective material compositions.

6.2. Alternative GPR Models

The use of different models of GPR with a variety of frequencies may improve the accurate detection and depth of cracks, including smaller cracks that are not visible. Models of higher resolution may be able to address the current incapability in detecting minute cracks that are not easily seen by the current GPR.

6.3. Use of other NDT Techniques

The results produced by GPR may be corroborated by the other supplementary NDT techniques like UPV. The same could be helpful in determining micro-pores or internal discontinuities within the concrete, that helps better for understanding moisture seepage characteristics.

6.4. Improve Processing Algorithms for Data

It may be that more sophisticated algorithms in software used like RADAN can clarify data in noisier or moisture-related areas. Signal processing, optimized for types of concrete used in IITJ buildings, could also result in sharper, more interpretable renderings.

6.5. Seasonal Variability

Seasonal studies on GPR surveys conducted in various seasons, especially during the monsoon and after, may reflect changes in absorption and retention of water upon the concrete and its impact upon its integrity over time. This would help in a seasonal comparison and further clarify environmental aspects that play a role in penetrating moisture and developing cracks.

6.6. Development of Protocol for Standardized Testing

The design-based protocol of deployment, data acquisition, and data processing should improve the reproducibility and can serve as a guideline for future campus-wide structural evaluation.

This set of recommendations shall be taken forward to further envisage and design a usable framework that can be further developed to detect hairline cracks in high-confined concrete structures and such problems related to moisture.

7. References

[1]. Daniels, D. J. (2004). *Ground Penetrating Radar*. 2nd ed. IET.

[2]. Mazzarella, M., et al. (2010). "GPR Investigation of Concrete Structures." *Journal of Applied Geophysics*.

[3]. Jol, H. M. (2009). *Ground Penetrating Radar: Theory and Applications*. Elsevier.

[4]. Kachroo, R., Bhat, M. (2016). "GPR for Crack Detection in Concrete Structures." *Concrete International*.

[5]. Müller, M., et al. (2015). "The Role of Early Detection in Concrete Crack Monitoring." *Concrete Research Journal*.

[6]. Andro, S., et al. (2012). "Optimal GPR Parameters for Concrete Crack Detection." *Journal of Nondestructive Evaluation*.

[7]. Huang, Z., et al. (2018). "Improving GPR Depth Calibration for Concrete Inspections." *Journal of Civil Engineering*.

[8]. Rahman, M. T., Hassan, M. (2013). "GPR Technology for Concrete Structural Health Monitoring." *Journal of Structural Engineering*.

[9]. Liu, Q., et al. (2015). "Application of GPR Migration in Concrete Slab Inspections." *Geophysical Journal International*.

[10]. Casini, L., et al. (2019). "Limitations of GPR in Small Crack Detection in Concrete." *Construction and Building Materials*.

[11]. Han, X., et al. (2020). "Integration of GPR and Machine Learning for Concrete Crack Detection." *Automation in Construction*.

[12] Maruddani, B., & Sandi, E. (2019). The development of Ground Penetrating Radar (GPR) data processing. *International Journal of Machine Learning and Computing*, *9*(6), 768–773.

[13] Acw22admin. (2023, November 14). Re: Ground Penetrating Radar (GPR) survey report GNO Bridge, *New Orleans - Southern Inspection and Testing, Inc. Southern Inspection and Testing,*